Unity 3D AR Restaurant Menu

Project Overview

The **AR Restaurant Menu** is an interactive Augmented Reality application developed using **Unity 3D**. The application allows customers to scan a restaurant menu or designated AR target and view realistic **3D food models** directly in their real-world environment.

The project is designed to provide an engaging and modern restaurant ordering experience by combining **Augmented Reality, 3D visualization, interactive UI, and mobile technology**.

Technologies Used

- **Unity 3D**
- **C# Programming**
- **Vuforia Engine**
- **Augmented Reality (AR)**
- **Vuforia AR Plugins**
- **3D Food Models**
- **Unity UI System**
- **Android SDK**
- **Java / JDK**
- **Gradle**
- **Android Build Tools**

Key Features

- AR-based restaurant menu system
- Image-target recognition using Vuforia Engine
- Real-time placement of 3D food items
- Interactive 3D food models
- Food model rotation and scaling
- Menu and food item selection
- Food information and descriptions
- Interactive buttons and user interface
- Restaurant-themed AR experience
- Camera-based AR interaction
- Mobile-friendly interface
- Multiple food/menu items can be displayed
- Smooth interaction with 3D objects
- Android mobile application support

AR Functionality

Vuforia Engine is integrated into Unity to detect predefined image targets. Once a target is recognized through the mobile camera, the corresponding **3D food model is displayed on top of the target**, creating an Augmented Reality experience.

Users can interact with the displayed food models and explore the restaurant's menu in an immersive way.

3D Visualization

The application uses optimized **3D food assets** to provide customers with a visual representation of dishes before ordering. Users can view dishes from different angles and interact with the models using touch controls.

User Interface

The application includes an interactive UI for:

- Viewing available dishes
- Selecting food items
- Displaying food names
- Providing food descriptions
- Navigating between menu sections
- Controlling AR interactions

Android Application

The Unity project is configured and built for **Android devices**. Android build configuration, SDK tools, JDK, Gradle, and required Unity/Vuforia components are used to generate the mobile application.

The final application can be installed on a compatible Android smartphone and used through the device camera.

Project Objective

The main objective of this project is to create a **modern AR-powered restaurant menu** that improves the traditional menu experience. Instead of viewing only images of dishes, customers can see interactive 3D representations of food in their real environment.

Final Deliverable

The completed project consists of a **Unity 3D AR Restaurant Menu application**, integrated with **Vuforia Engine and AR functionality**, with an Android build that allows users to experience interactive 3D restaurant dishes through their smartphone camera.